Problem-Solution Fit

Date	05 July 2026
Team ID	SWTID-2026-8308
Project Name	My project is opticrop: smart agricultureal production optimization engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Small and medium-scale farmers - Large commercial farms - Agricultural cooperatives - Agri-business companies - Government agriculture departments	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** - Limited budget - Low digital literacy - Poor internet connectivity - Lack of smart devices and sensors - Limited access to modern farming technologies	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** **Existing Solutions:** - Traditional farming methods - Weather forecast apps - Basic farm management software **Pros:** - Easy to use - Low initial cost - Basic information available **Cons:** - No AI-based recommendations - Limited data integration - Low prediction accuracy - No real-time optimization
2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]** - Unpredictable weather conditions (High) - Low crop yield due to poor planning (High) - Excessive water and fertilizer usage (High) - Pest and disease outbreaks (Medium) - Unstable market prices (Medium)	**9. PROBLEM ROOT / CAUSE [RC]** - Lack of real-time agricultural insights - Inefficient resource management - Climate variability - Limited access to accurate farming data - Poor decision-making due to insufficient analytics	**7. BEHAVIOR** + ITS INTENSITY **[BE]** - Farmers rely on traditional practices (High) - Decisions based on experience rather than data (High) - Technology adoption is gradually increasing (Medium) - Need quick and simple recommendations (High)

3. TRIGGERS TO ACT [TR] - Unpredictable weather conditions - Declining crop yield - Rising farming costs - Water scarcity - Pest and disease outbreaks - Need for higher profits and sustainable farming	**10. YOUR SOLUTION [SL]** **OptiCrop** is an AI-powered smart agricultural production optimization engine that analyzes soil health, weather forecasts, crop conditions, and market trends to provide personalized recommendations for crop selection, irrigation scheduling, fertilizer usage, pest detection, and yield prediction. It helps farmers improve productivity, reduce costs, conserve resources, and make datadriven farming decisions.	**8. CHANNELS OF BEHAVIOR [CH]** ONLINE - Mobile application - Web dashboard - SMS/WhatsApp alerts - Agricultural websites - Social media (YouTube, Facebook)
4. EMOTIONS BEFORE / AFTER **[EM]** **Before:** - Uncertainty about farming		OFFLINE - Agricultural extension officers - Farmer meetings and workshops

decisions - Stress due to crop failures - Financial insecurity - Fear of unpredictable weather **After:** - Confidence in decisionmaking - Increased productivity - Better income and profitability - Satisfaction with optimized resource usage		- Local agricultural cooperatives - Government agriculture offices - Training programs and field demonstrations